7.5 Solving Rational Equations

Name _____ Period ____ Date _____

Algebra II 7.5 (30-minute plan) · Student sheet

I can…

- solve rational equations by cross multiplying.
- solve rational equations using the LCD and check for extraneous solutions.
- use rational equations to solve real-life problems.

Vocabulary · fill in the missing word

Rational equation — an equation that contains rational _____

Cross multiplying — if $\frac{a}{b}$ = _____, then ad = bc

Extraneous solution — a solution of a later step that is not a solution of the _____ equation (it makes a denominator 0)

Part 1 · Notes and examples (fill in each note, then work the example)

Rational equations: One fraction on each side: _____.

Example 1

Solve $\frac{3}{x+1} = \frac{9}{4x+5}$.

More fractions: More fractions: multiply _____ term by the LCD.

Example 2

Solve $\frac{5}{x} + \frac{7}{4} = -\frac{9}{x}$.

Extraneous solutions: An **extraneous solution** makes a _____ equal 0. Always _____ in the original equation.

Example 3

Solve $\frac{x}{x-2} = \frac{2}{x-2} + 3$.

Part 2 · Partner practice

Partner practice

Solve $\frac{4}{x-2} = \frac{6}{x}$.